ADVANCED EXCEL PROJECT • PROJECT SYNOPSIS

Smart City Operations Dashboard

Transport • Energy • Waste • Emergency Services

A 30-day, five-zone simulated operations-centre view built in Microsoft Excel

68,148	37,128.5	4,864.3	228
passengers	MWh energy	tonnes waste	incidents

Source: the supplied presentation “smart_city_ecommerce_template_blue_green.pptx” (10 slides).

All figures come from the supplied 30-day dataset; this is not live city telemetry.

Prepared 9 October 2026

02 Executive Summary

What the project is, what it shows and why it matters

The Smart City Operations Dashboard is an Advanced Excel project that simulates a city control centre. It brings four critical operating domains – transport, energy, waste management and emergency services – into a single interactive workbook, so that raw operational records can be turned into clear measures and, from those, into decisions.

The workbook is fed by four raw-data sheets, each paired with a calculation sheet, and a dashboard that sits on top. It covers five city zones over a 30-day period. Excel techniques such as XLOOKUP, INDEX-MATCH, conditional aggregation, PivotTables and slicers drive the analysis, and a quality-check routine verifies the data before any insight is presented.

Headline results for the 30 days:

TRANSPORT
68,148
passengers; 4.6 min average delay

ENERGY
37,128.5 MWh
12 outages in total

WASTE
4,864.3 tonnes
57.2% recycling rate

EMERGENCY
228 incidents
12.3 min average response

Project at a glance

Item	Detail
Project type	Advanced Excel project: interactive multi-domain operations dashboard
Domains	Transport, Energy, Waste, Emergency Services
Coverage	5 city zones (A–E), 30 days of operational data
Workbook structure	4 raw-data sheets + 4 calculation sheets + dashboard + formula examples
Core techniques	XLOOKUP, INDEX-MATCH, SUMIFS / COUNTIFS / AVERAGEIFS, IF + COUNTIF, MAXIFS, PivotTables and slicers
Data nature	Supplied simulated dataset, not live telemetry

The five decisions the dashboard points to are covered in section 9: solar expansion for Zone C, a scheduling review for Route 102, a 90% on-time target for recyclable waste, extra response units for Zone E, and overload monitoring for Zone B.

03 Project Purpose, Scope and Objectives

Why the project was built and what it sets out to demonstrate

Purpose

City operators rarely look at one service in isolation. A late bus, a busy power grid, a missed waste collection and a slow emergency response all affect the same residents. The project therefore treats the city as one operations centre: a simulated control room where the four domains can be monitored side by side and compared.

Objectives

- Demonstrate XLOOKUP, INDEX-MATCH and conditional aggregation on realistic multi-sheet data.
- Build an interactive, multi-domain operational dashboard.
- Use filtering, PivotTables and visual status indicators so that the user can explore the data.
- Turn raw city data into decisions for transport, energy, waste and emergency services.

Operations lens: what each domain measures

Domain	Focus of the analysis	Headline measures (30 days)
Transport	Passengers, delays and route performance	68,148 passengers; 4.6 min average delay
Energy	Consumption, outages and grid load	37,128.5 MWh; 12 outages
Waste	Waste volume and recycling rate	4,864.3 tonnes; 57.2% recycling rate
Emergency	Incident volume and response time	228 incidents; 12.3 min average response

Scope and boundaries

- In scope: five city zones, 30 days of data, four raw-data streams and their calculation layers, a dashboard and example formulas.
- Deliverables named in the project: the Excel workbook plus professional PDF and PPT documentation.
- Out of scope: live data feeds. The project states clearly that its insights come from the supplied dataset and are not real-time city telemetry.

04 Dataset and Data Model

Four operational data streams feed one dashboard

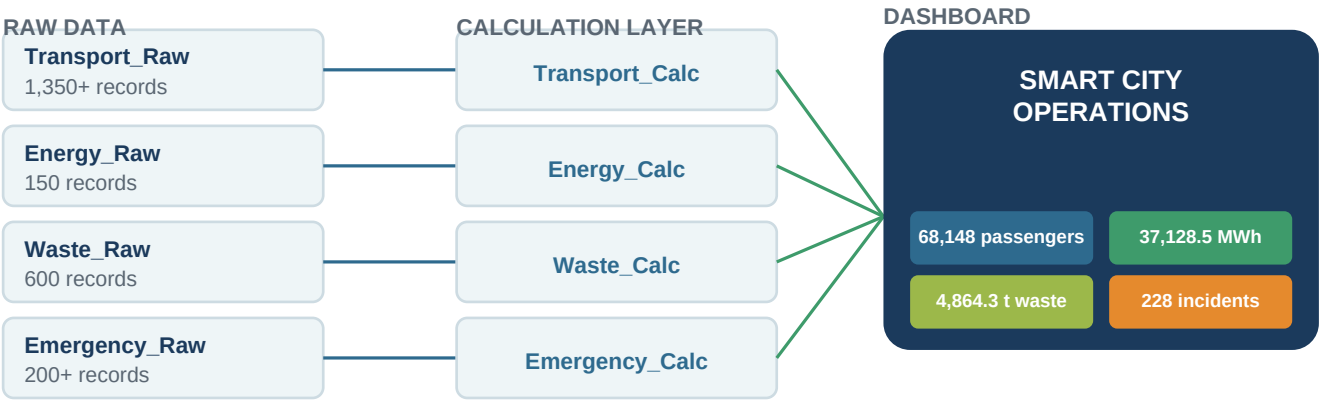


Figure: the three-layer workbook structure described in the project (raw data → calculation → dashboard).

How the layers work

Raw layer. Each domain has its own untouched data sheet: Transport_Raw (1,350+ records), Energy_Raw (150 records), Waste_Raw (600 records) and Emergency_Raw (200+ records). Keeping the raw data separate means the source records are never overwritten by calculations.

Calculation layer. A matching sheet for each domain (Transport_Calc, Energy_Calc, Waste_Calc, Emergency_Calc) holds the formulas – lookups and conditional aggregations – that summarise the raw records by zone, route, waste type or incident type.

Dashboard layer. The dashboard reads from the calculation sheets and presents the headline measures and charts in one place.

Property	Value
Raw-data sheets	4 (transport, energy, waste, emergency)
Calculation sheets	4 (one per domain)
Other sheets	Dashboard and formula examples
City zones	5 (Zones A–E)
Time period	30 days of operational data
Records	Transport 1,350+; Energy 150; Waste 600; Emergency 200+

05 Excel Approach

Core techniques used to turn city data into operational insights

The project is built to show a practical toolkit rather than a single trick. Six techniques are combined, each solving a different part of the reporting problem.

Technique	What it is used for in the project	Why it matters
XLOOKUP	Looks up zone-level energy totals	Fetches a value by key without fixed column positions
INDEX-MATCH	Multi-criteria lookups	Flexible retrieval when more than one condition identifies a row
SUMIFS / COUNTIFS / AVERAGEIFS	Conditional aggregation	Totals, counts and averages per zone, route, type or status
IF + COUNTIF	Visual status indicators	Turns a number into a clear flag the operator can read at a glance
MAXIFS	Finds maximum values	Surfaces the worst or highest case, such as the longest delay
PivotTables + Slicers	Interactive filtering	Lets the user slice the data by zone or category without editing formulas

Quality checks before presenting insights

The project does not present figures until the workbook has been verified. The checks named in the presentation are:

- **Key uniqueness:** identifiers that should be unique are confirmed unique.
- **Nulls:** missing values are found and handled rather than silently ignored.
- **Numeric types:** numbers stored as text are caught, since they break sums and averages.
- **Lookup consistency:** lookups return the right match and do not fail quietly.
- **Dashboard totals:** headline totals are reconciled against the underlying data.

Why this combination works

Lookup functions retrieve data from the right place, conditional aggregation condenses hundreds of rows into a few meaningful numbers, status indicators make those numbers readable, and PivotTables with slicers let the user explore further. The quality checks then protect the credibility of everything above them.

06 Executive Dashboard

30-day headline performance metrics

The executive dashboard presents eight headline measures, two for each domain, so that operators can read volume and performance at a glance.

TOTAL PASSENGERS **68,148** transport	AVERAGE DELAY **4.6 min** transport	TOTAL ENERGY **37,128.5 MWh** energy	TOTAL OUTAGES **12** energy
TOTAL WASTE **4,864.3 tonnes** waste	RECYCLING RATE **57.2%** waste	TOTAL INCIDENTS **228** emergency	AVG RESPONSE TIME **12.3 min** emergency

Reading the numbers

Measure	Value	What it tells the operator
Total passengers	68,148	Overall demand placed on the transport network over 30 days
Average delay	4.6 min	Typical lateness per service; a service-quality indicator
Total energy	37,128.5 MWh	Overall consumption across the five zones
Total outages	12	How often supply was interrupted; a reliability indicator
Total waste	4,864.3 tonnes	Volume of waste handled in the period
Recycling rate	57.2%	Share of waste recycled; a sustainability indicator
Total incidents	228	Volume of emergency calls handled
Avg response time	12.3 min	Typical time to respond; a public-safety indicator

Volume measures (passengers, energy, waste, incidents) show how much work each service carries. Performance measures (delay, outages, recycling rate, response time) show how well it is doing. Together they give the “service pressure” view that the dashboard is designed to provide.

07 Operational Performance by Domain

How the four city domains contribute to the 30-day picture

Transport

68,148 passengers | 4.6 min average delay

Transport is measured by how many passengers travel and how late services run. The graphical analysis breaks bus performance down by route (Routes 101, 102, 103, 201, 202 and 301), which is what allows a single problem route to be identified.

Energy

37,128.5 MWh | 12 outages

Energy is tracked through consumption, outages and grid load. The analysis compares average solar output and average outages for each of the five zones, and the insights note which zones draw the most energy and carry the highest grid load.

Waste

4,864.3 tonnes | 57.2% recycling rate

Waste is measured by volume and by recycling performance. The analysis splits collections into on-time, delayed and missed for each zone across four waste types: e-waste, hazardous, organic and recyclable.

Emergency

228 incidents | 12.3 min average response

Emergency services are measured by incident volume and response time. The analysis totals incidents by type – fire, medical, traffic accident, utility failure and other – across the zones, and flags where critical incidents take longest to answer.

The dashboard is designed as an operations-centre view: each domain has its own clear set of measures, while the workbook supports analysis across domains.

08 Graphical Analysis

The four charts that accompany the headline measures

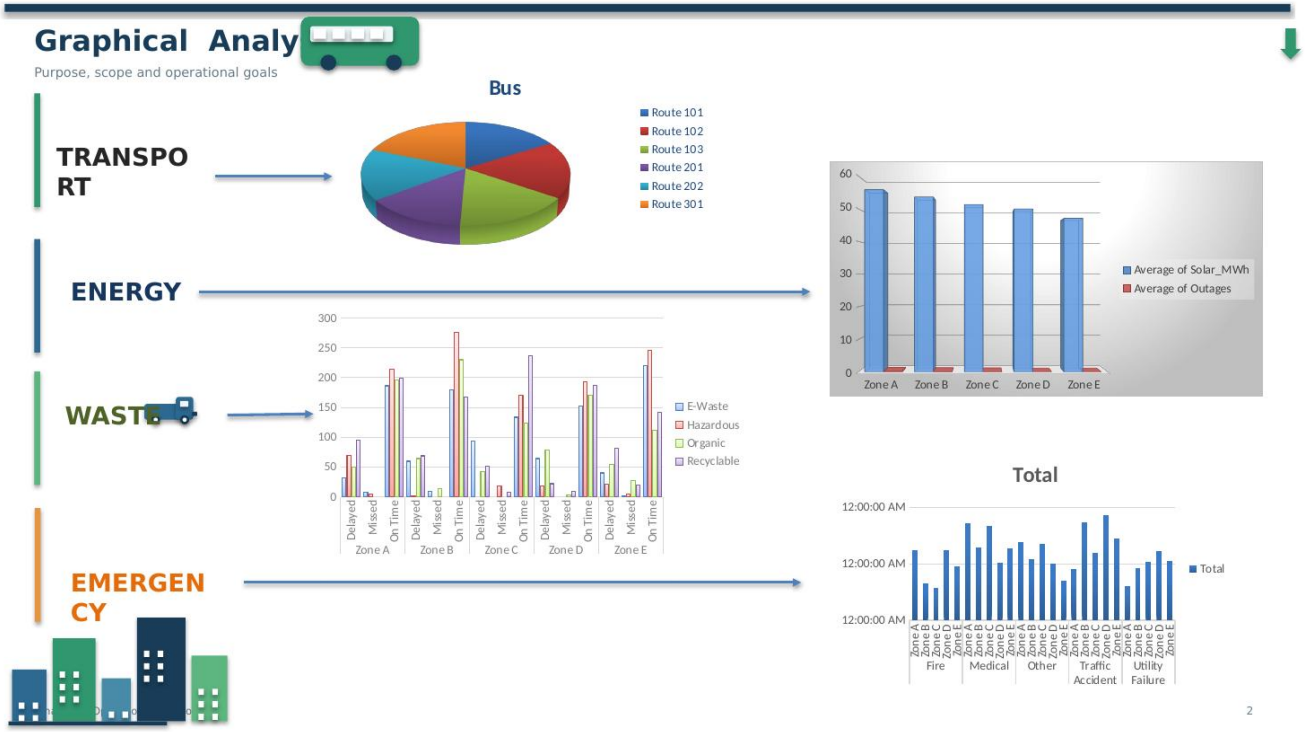


Figure: the project's "Graphical Analysis" slide, showing one chart for each domain.

Domain	Chart	What it shows
Transport	3-D pie chart, "Bus"	Share of the bus data by route: Routes 101, 102, 103, 201, 202 and 301
Energy	Column chart	Average solar output (MWh) and average outages for Zones A to E; solar averages fall from about 55 MWh in Zone A to about 47 MWh in Zone E, while average outages are very small by comparison
Waste	Clustered column chart	On-time, delayed and missed collections for each zone, split by e-waste, hazardous, organic and recyclable waste; on-time collections are by far the largest group
Emergency	Column chart, "Total"	Totals by incident type (fire, medical, other, traffic accident, utility failure) and zone

Chart observations

- The charts give a quick visual comparison across zones and categories, and they support the specific insights on the next page.
- On the emergency chart the value axis is labelled with time-style values ("12:00:00 AM"), so the axis format should be changed to a number or minutes format before the dashboard is relied on.
- The slide title is partly covered by the bus icon, and the "Transport" and "Emergency" labels wrap awkwardly. These are cosmetic presentation fixes.

09 Key Insights and Decisions

What the dashboard reveals – and the operational response it suggests

The dashboard is only useful if it leads to action. The project turns its findings into five specific operational responses, one or more for each domain.

Area	Finding	Suggested response
Zone C (energy)	Highest energy consumption	Consider solar expansion
Route 102 (transport)	Highest transport delays	Investigate scheduling
Recyclable waste	75% on-time rate	Target 90%
Zone E (emergency)	Critical incidents average 15+ minutes to respond	Add response units
Zone B (energy)	Grid load averages 65%	Monitor overload risk

How to read each decision

Zone C – solar expansion. The zone that uses the most energy is the one where additional local generation would have the greatest effect on demand from the grid.

Route 102 – scheduling. Delays on one route are usually a timetable, spacing or congestion issue; reviewing the schedule is the lowest-cost first step before adding vehicles.

Recyclable waste – 90% target. At 75% on-time, recyclable collection trails the aim. Setting an explicit 90% target gives operators a measurable goal to track on the dashboard.

Zone E – response units. Critical incidents averaging more than 15 minutes is well above the city-wide average of 12.3 minutes, which points to a capacity gap in that zone.

Zone B – overload monitoring. A grid load that averages 65% leaves limited headroom for peaks, so the zone should be watched for overload risk.

Important caveat: these are portfolio-level insights from the supplied 30-day dataset. They are not live city telemetry and should not be treated as real operating decisions.

10 Learning Outcomes and Conclusion

Skills demonstrated and where the project goes next

Skill area	What the project demonstrates
Lookup engineering	XLOOKUP and INDEX-MATCH for zone-level and multi-criteria retrieval
Conditional analysis	SUMIFS, COUNTIFS and AVERAGEIFS for operational metrics
Interactive reporting	PivotTables, slicers, KPIs and charts for dashboard use
Data modelling	Separate raw-data and calculation layers across domains
Documentation	Professional Excel, PDF and PPT deliverables

Conclusion

The project shows how four operational domains can be combined into one clear, interactive view of city performance. Its approach follows four steps: **connect** – structure raw data across the four domains; **calculate** – use lookups and conditional aggregation; **visualise** – build KPIs, PivotTables and interactive views; and **act** – translate the patterns into operational decisions.

It is a portfolio-ready example of how spreadsheet engineering can support an operations dashboard.

Limitations

- The data is a supplied simulated dataset covering 30 days, so results cannot be assumed to hold over a longer period or for a real city.
- The presentation reports headline values and chart visuals only. The underlying workbook was not part of the material supplied, so these figures could not be recalculated or checked independently.
- Some chart formatting needs tidying (the emergency-chart axis format and a few slide labels).

Possible next steps

- Extend the dataset beyond 30 days to test whether the zone and route patterns persist.
- Add cost and capacity data so that proposals such as solar expansion or extra response units can be compared on value.
- Set alert thresholds in the dashboard, for example a response-time limit or a grid-load ceiling, using the existing status indicators.
- Connect the workbook to a live or regularly refreshed data source if it is to move beyond a simulation.